

ELITE IGCSE MATHEMATICS
EXPERIENCE NOTES

Percentages

Learn the trigger, write the first line, then solve one similar problem while the method is still fresh.

PERCENTAGES · CH1 · L6

ENG. ESLAM AHMED



The Map

This lesson collapses Percentages into the few exam moves that keep repeating. Read the trigger, copy the first line, then practise the same idea on the next page.

01 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

02 Reverse a percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

03 Use compound change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

04 Compare percentage change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

05 Calculate percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

06 Percentage profit

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BANK EVIDENCE 40 website questions, 40 checked solutions, 3 local PDFs read.

01 Use percentage multiplier

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$100\% - 15\% = 85\%$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P1HR Q7

The hat is reduced by

$$100\% - 15\% = 85\%$$

of the normal price

$$0.85 \times \text{normal price} = 20.40$$

Finish: 24 euros.

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times (1 + \frac{r}{100})$, decrease means $\times (1 - \frac{r}{100})$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

SAME IDEA Use the same first line from Strategy 01.

QUESTION

7 In a sale, the normal price of a hat is reduced by 15% The sale price of the hat is 20.40 euros. Work out the normal price of the hat. euros

YOUR SOLUTION

02 Reverse a percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$100\% - 15\% = 85\%$$

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SAME IDEA Use the same first line from Strategy 02.

QUESTION

7 In a sale, the normal price of a hat is reduced by 15% The sale price of the hat is 20.40 euros. Work out the normal price of the hat. euros

YOUR SOLUTION

03 Use compound change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$14 \times 7 \times 5 = 490$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2HR Q6

Pierre's old weekly pay

This is a reasoning step: write one clear sentence, then continue the calculation.

His new weekly pay is

$$14 \times 7 \times 5 = 490$$

Finish: Pierre's pay increased by 22.5%. Marie's old salary was 24500 euros.

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times (1 + \frac{r}{100})$, decrease means $\times (1 - \frac{r}{100})$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 03

SAME IDEA Use the same first line from Strategy 03.

QUESTION

6 In his previous job, Pierre was paid 400 euros in total for working a 5-day week. In his new job, Pierre is paid 14 euros per hour. In his new job, Pierre works for 7 hours each day for a 5-day week. (a) Work out the percentage increase in the amount that Pierre is paid for a 5-day week. % (4) Marie changes her job. Her salary decreases by 6% Her new salary is 23 030 euros. (b) Work out Marie's salary before she changes her job. euros (3)

YOUR SOLUTION

04 Compare percentage change

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$14 \times 7 \times 5 = 490$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jun 2023 P2HR Q6

Pierre's old weekly pay

This is a reasoning step: write one clear sentence, then continue the calculation.

His new weekly pay is

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Finish: Pierre's pay increased by 22.5%. Marie's old salary was 24500 euros.

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times (1 + \frac{r}{100})$, decrease means $\times (1 - \frac{r}{100})$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 04

SAME IDEA Use the same first line from Strategy 04.

QUESTION

6 In his previous job, Pierre was paid 400 euros in total for working a 5-day week. In his new job, Pierre is paid 14 euros per hour. In his new job, Pierre works for 7 hours each day for a 5-day week. (a) Work out the percentage increase in the amount that Pierre is paid for a 5-day week. % (4) Marie changes her job. Her salary decreases by 6% Her new salary is 23 030 euros. (b) Work out Marie's salary before she changes her job. euros (3)

YOUR SOLUTION

05 Calculate percentage

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$545 - 500 = 45$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2020 P2H Q8

Convert standard form

$$545 - 500 = 45$$

Convert standard form

$$592 - 545 = 47$$

Finish: Brigitte is not correct; the discount is 168 euros.

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times(1 + \frac{r}{100})$, decrease means $\times(1 - \frac{r}{100})$.

— BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 05

SAME IDEA Use the same first line from Strategy 05.

QUESTION

8 The table shows the cost, in euros, of Brigitte's car insurance in each of the years 2016, 2017 and 2018

Year	2016	2017	2018
Cost of insurance (euros)	500	545	592

Brigitte says, "The percentage increase in the cost of my car insurance from 2017 to 2018 is more than the percentage increase in the cost of my car insurance from 2016 to 2017"

(a) Is Brigitte correct? You must show how you get your answer. (4)

Henri wants to insure his car. He gets a discount of 15% off the normal price. Henri pays 952 euros for his ...

YOUR SOLUTION

06 Percentage profit

TRIGGER *"work out", "show", "hence", a familiar exam phrase*

BECOMES A Percentages question where the mark is won by naming the move first, then substituting cleanly.

FIRST LINE TO WRITE

$$520 - 465 = 55$$

SIMPLEST STRATEGY

- 1 Underline the target: what is the question asking you to find?
- 2 Write the rule, theorem, or equation before any calculator work.
- 3 Substitute one value at a time and keep units/unknowns visible.
- 4 Answer in the exact form requested: units, degree, money, ratio, or proof reason.

WORKED MODEL

Source pattern: Jan 2021 P2H Q3

Profit

$$520 - 465 = 55$$

Percentage profit

$$\frac{55}{465} \times 100 = 11.827 \dots$$

Finish: (a) 11.8%, (b) \$484

FORMULA BANK

Percentage multiplier: increase by $r\%$ means $\times (1 + \frac{r}{100})$, decrease means $\times (1 - \frac{r}{100})$.

BOTTOM LINE

The examiner is not looking for magic; they are looking for the correct first line, clean substitution, and the requested final form.

Practice 06

SAME IDEA Use the same first line from Strategy 06.

QUESTION

3 Gladys buys a table for \$465 to sell in her shop. She sells the table for \$520 (a) Work out the percentage profit that Gladys makes from the sale of the table. Give your answer correct to 3 significant figures. % (3) Gladys has a sale in her shop. She decreases all the normal prices by 12% The normal price of an armchair was \$550 (b) Work out the sale price of the armchair. \$ (3)

YOUR SOLUTION

Quick Reference

TRIGGER → FIRST MOVE

WHEN YOU SEE	WRITE THIS FIRST
"work out", "show", "hence", a familiar exam phrase	Use percentage multiplier
"work out", "show", "hence", a familiar exam phrase	Reverse a percentage
"work out", "show", "hence", a familiar exam phrase	Use compound change
"work out", "show", "hence", a familiar exam phrase	Compare percentage change
"work out", "show", "hence", a familiar exam phrase	Calculate percentage
"work out", "show", "hence", a familiar exam phrase	Percentage profit

FINAL BOTTOM LINE Do not try to remember every past-paper wording. Remember the trigger, write the first line, and let the working follow.